What is Clustering?

Clustering is an **unsupervised learning** technique where the model groups similar data points together **without any labels.**

No one tells the model what the groups are. It finds the natural structure in data by itself.

Simple example: You have 10,000 customers. You don't know their categories. Clustering will automatically group them — budget customers, premium customers, occasional buyers — without you defining these groups.

Where Clustering is Used in Real Life

- **E-commerce** — group customers by purchase behavior
- **Healthcare** — group patients by symptoms
- **Marketing** — segment audience for targeted ads
- **Cybersecurity** — detect unusual network behavior
- **Document analysis** — group similar news articles
- **Genetics** — group genes with similar expression patterns

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Types of Clustering Algorithms

1️⃣ K-Means Clustering

Most popular and widely used clustering algorithm.

Simple idea: Divide data into K groups where each point belongs to the group with the nearest center.

How it works step by step:

1. Choose K — number of clusters you want
2. Randomly place K centroids in the data
3. Assign each data point to nearest centroid
4. Recalculate centroid as mean of all points in that cluster
5. Repeat steps 3 and 4 until centroids stop moving
6. Final clusters are the result

Real example: Group 1000 students into 3 groups based on study hours and marks. K-Means will find natural groupings — high performers, average, low performers.

Best for:

- Large datasets
- When you know number of clusters
- Spherical shaped clusters
- Simple and fast

Weakness:

- You must define K in advance — how many clusters?
- Sensitive to outliers
- Assumes clusters are spherical and equal size
- Different runs can give different results

How to choose K — Elbow Method: Run K-Means with different values of K. Plot the error for each K. The point where error stops decreasing sharply — that is the best K. It looks like an elbow on the graph.

2️⃣ Hierarchical Clustering

Builds a **tree of clusters** called a **dendrogram.** You don't need to specify K in advance.

Two approaches:

Agglomerative — bottom up:

- Start with each point as its own cluster
- Merge the two closest clusters
- Repeat until everything is one cluster
- Cut the dendrogram at desired level to get clusters

Divisive — top down:

- Start with all points in one cluster
- Split into two clusters
- Repeat until each point is its own cluster

Best for:

- When you don't know number of clusters
- Small to medium datasets
- When you want to visualize cluster relationships

Weakness:

- Slow on large datasets
- Once merged or split — cannot undo

3️⃣ DBSCAN — Density Based Spatial Clustering

Groups points that are **closely packed together** and marks points in low density regions as outliers.

Think of it like finding crowded areas in a city. Dense areas are clusters. Isolated areas are noise.

Key concepts:

- **Core point** — has minimum number of neighbors within radius
- **Border point** — near a core point but not enough neighbors
- **Noise point** — isolated point, not part of any cluster — outlier

Best for: When clusters have irregular shapes

- When data has outliers — DBSCAN automatically detects them
- When you don't know number of clusters

Weakness:

- Choosing radius and minimum points is tricky
- Struggles with varying density clusters

K-Means vs Hierarchical vs DBSCAN

	K-Means	**Hierarchical**	**DBSCAN**
Need K?	Yes	No	No
Cluster shape	Spherical	Any	Any
Outliers	Sensitive	Sensitive	Detects automatically
Dataset size	Large	Small/Medium	Medium
Speed	Fast	Slow	Medium

Evaluation Metrics for Clustering

Clustering is harder to evaluate because there are no labels. But we have some metrics:

Inertia: Sum of distances between each point and its cluster center. Lower is better. Used in Elbow method.

Silhouette Score: Measures how similar a point is to its own cluster compared to other clusters. Score between -1 and 1. Closer to 1 is better.

Davies-Bouldin Index: Lower score means better defined clusters.

Practice These

What is clustering?

"Clustering is an unsupervised learning technique that groups similar data points together without any predefined labels. The model finds natural structure in data by itself. Common algorithms include K-Means, Hierarchical Clustering and DBSCAN."

How does K-Means work?

"K-Means starts by randomly placing K centroids. It assigns each data point to the nearest centroid, then recalculates centroids as the mean of all points in that cluster. This repeats until centroids stop moving. The final clusters are the result."

How do you choose K in K-Means?

"We use the Elbow Method. We run K-Means with different values of K and plot the inertia for each. The point where inertia stops decreasing sharply looks like an elbow on the graph — that is the optimal K."

What is the difference between K-Means and DBSCAN?

"K-Means requires you to specify the number of clusters in advance and assumes spherical cluster shapes. DBSCAN does not need the number of clusters — it finds clusters based on density and automatically identifies outliers as noise points. DBSCAN works better for irregular shaped clusters."